- If you plan to use a longer power cable, see *Control Module Connections* on page 5-53.

Step 6. To mount the Control Module, place it onto its mounting bracket so that its grove at the bottom (2) rests on the lip at the bottom of its mounting bracket (2).

Step 7. Press the Control Module back onto the mounting bracket such that the top of the Control Module fits under the mounting bracket's tabs (3) at the top. The screw holes (3) on the top of the Control Module should line up with the mounting brackets tabs (3).

5.8.2 Control Module Connections

The Control Module includes the following connections:

- Power Cable
- External On/Off Signal
- General Alarm Signal
- Data Cable
- Modbus

Avoid direct contact with the electronic circuit board which is sensitive to electrostatic discharge (ESD). Utilize best practices regarding ESD when replacing the power cable.

➲ To connect peripheral equipment:

Opening the Control Module's Back

Step 1. Using the appropriate screwdriver, unfasten the screws on the Control Module (A) and carefully remove the back.

Figure 5-18: Control Module's Back

Connecting the Signals

Step 2. To connect an **External On/Off** signal, thread the signal cable through one of the holes with plastic glands on the bottom (5) and connect the wires to the appropriate connectors as show to the right (6) and as shown in **Figure** 5-20 as terminals **4** and **5** (corresponding to **6** in **Figure** 5-18).

Step 3. To connect a.n **General Alarm** signal, thread the signal cable through one of the holes with plastic glands on the bottom (5) and connect the wires to the appropriate connectors as show to the right (6) and as shown in **Figure** 5-20 us terminals **6**, **7** and **8** (corresponding to **6** in **Figure** 5-18).

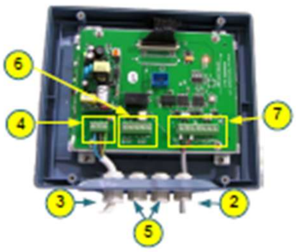

Figure 5-19: Control Module's Circuit Board

Connecting the Modbus

Step 4. To connect to an external facility control via Modbus, thread the signal cable through one of the holes with plastic glands on the bottom (5) and connect the wires to the appropriate connectors as show to the right (7) and as shown in **Figure** 5-20 as terminals **12**, **13**, **14** and **15** (corresponding to **6** in **Figure** 5-20). For more information, see **Appendix** A, ***Modbus Communication Protocol***. Add jumpers to any unused pins as shown in the diagram below.

Adding a Longer Data Cable

Step 5. To replace the Data cable with a longer one:

a Unfasten the Data cable from the connectors as show to the above (7) as terminals 9, 10 and 11 (- GND +) (corresponding to 6 in Figure 5-18).

b Loosen the plastic nut of gland (2) and pull it through and out of the plastic gland at the bottom.

c Thread the longer Data cable through its plastic gland (2) and fasten it to the appropriate connectors mentioned above.

5.8.3 PIN Assignments

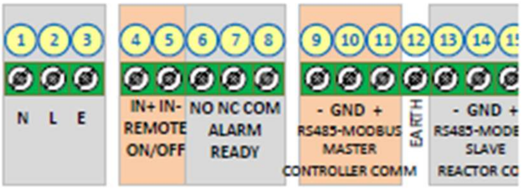

Figure 5-20: Control Module Terminal Connectors

Schematic Diagrams

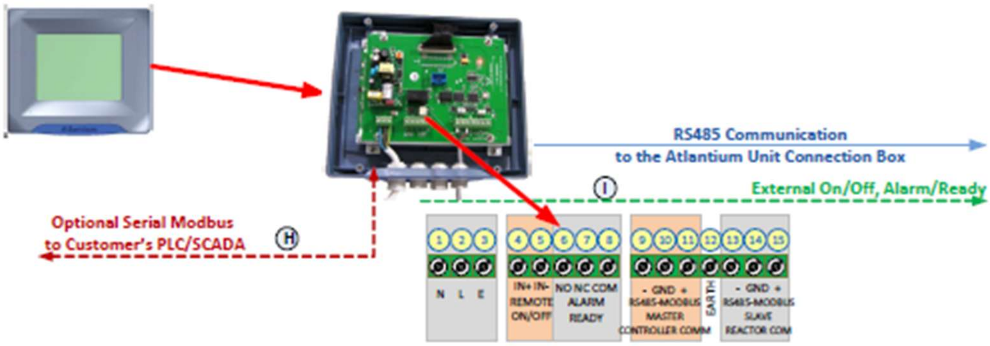

Figure 5-21: Control Module Terminal Connectors Schematic Drawing

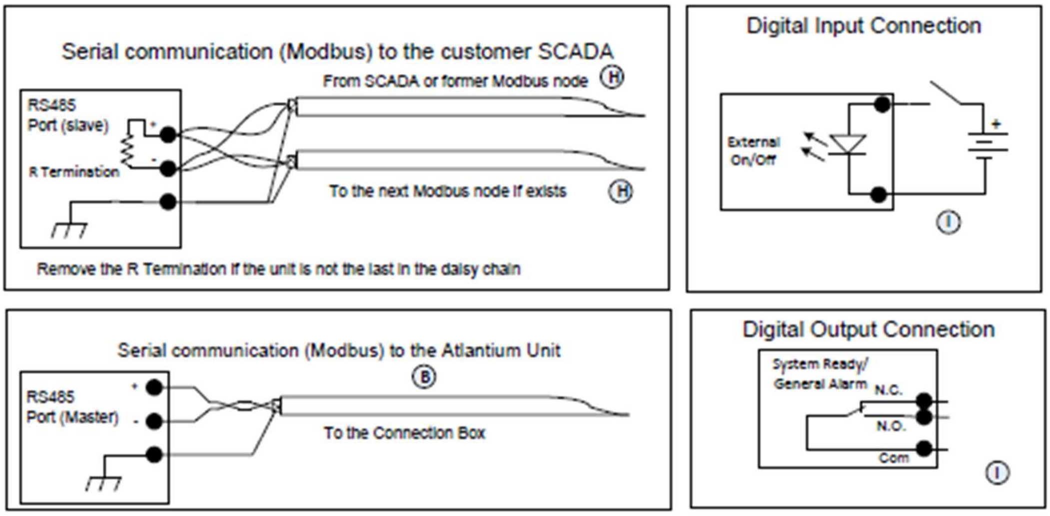

Figure 5-22: Control Module Terminal Connectors Wiring Drawings

Table 5-2: Peripheral Equipment Cable Details

#	Item	Sizes	Cat. No.	Description	Diameter	Supplied by
B.	Control Module RS485 data communication cable	5m/16.4ft. 10m/32.8ft.	HSL013200 HLS014500	Shielded twisted pair, 22AWG, 300V(General cables C0720A)	0.25 inch	Atlantium
G.	Flow Meter 4-20mA or Flow Switch cable Dry contact (See *Connection Box* on page 5-60.)	Customer defined	-	Connect to Connection Box no.1 through gland size 7. Wires size according to Flow Meter / Flow Switch manufacturer's instructions	0.25 inch	Customer
H.	RS485 Data communication cable	Customer defined	-	Shielded twisted pair, 22AWG, 300V	0.25 inch	Customer
I.	Digital output, Digital inputs, Analog outputs	Customer defined	-	Shielded cable, pairs of 22AWG, 300V	According to the number of pairs	Customer
R.	Laptop Control Module interconnect	Customer defined	-	Standard ETHERNET cable	-	Customer

Table 5-3: **Control Module Connector Pins**

#	Item	
1, 2, 3	Power cable connection.	
4	External On/Off IN+	On: 20?28VDC Off: <1VDC See *External On/Off - Additional Notes* below.
5	External On/Off IN-	
6	General Alarm/System Ready Dry Contact Normally Open **NO**	
7	General Alarm/System Ready Dry Contact Normally Closed **NC**	
8	General Alarm/System Ready Dry Contact Common **COM**	
	Data Cable Connectors	
9	RS485 / Modbus Master TX-	Cable is provided already attached. For information on attaching the cable to the Connection Box of Lamp 1. .
10	RS485 / Modbus Master Ground	
11	RS485 / Modbus Master TX+	
12	Earthing connector	
13	RS485 / Modbus Slave TX- to connect to an external control system	The Control Module is provided including a **100R, <5% , 0.25W, Through hole resistor** between #13 and #15. This resistor is to be removed in Control Modules connected by daisy-chain, but is to be left attached in the last Control Module of the daisy-chain.
14	RS485 / Modbus Slave Ground	
15	RS485 / Modbus Slave TX+	

5.8.4 External On/Off - Additional Notes

For regular wiring:

- Positive voltage must be connected to pin 4 – IN+
- Negative (GND) must be connected to pin 5 – IN-

Table 5-4: External On/Off

Control Module Screen More Settings	Physical Connection	Physical Connection	
	IN+	IN-	Lamp Control
NC Unchecked	0 VDC	0 VDC	Off
NC Unchecked	+24 VDC	0 VDC	On
NC Checked	0 VDC	0 VDC	On
NC Checked	+24 VDC	0 VDC	Off

6 Configuring the Control Module

6.1 Main Operation Screen

The Standard Control Module controls the operations and dose measurements of the Atlantium system.

The System uses a validated algorithm to measure the dose and display it in real time. Depending on how you choose to run your process, you can configure the Control Module to maintain a specific dose or to simply use a particular power level and inform you if it does not reach the dose you designate.

The Control Module provides a real-time display of operational data.

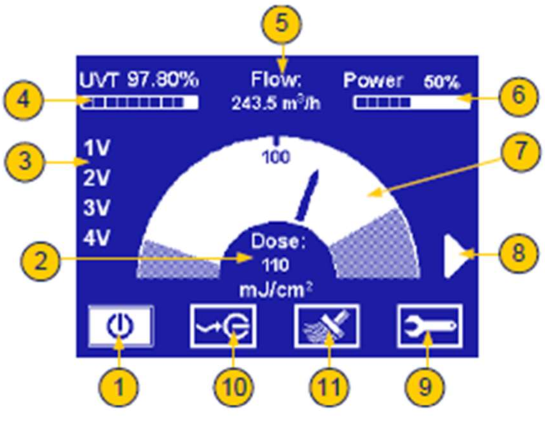

Figure 6-1: Initial Main Operation Screen

Table 6-1: Main screen components

#	Item
1	**Lamp Power Button** - upon Control Module initiation, is on the Off position. After being pressed, turns white, showing that the lamps are working. It can be used to shut lamps down in an orderly manner.
2	**Dose** - this is the dose the user wants the Unit to achieve and maintain. Generally, it should be set about 10-20% above the minimum required dose to assure stability and on-spec operation in the case of small variations of water quality or flow. You should confer with the Atlantium application engineer in setting this number based on the capacity of the Unit, variability at the site and power considerations. When the actual dose is below the Op(erate) At dose, the number rounds down. Otherwise, it rounds up.
3	Numbers 1-4 represents up to four lamps included in the System. V indicates lamps in operation (see Table 6-2)
4	**UVT** - scale indicates the current quality of the water flowing through the system, and the amount of the UV light absorbed by the water, relative to the intensity of UV light produced by the lamp. If the UVT percentage factor is high, the clarity factor of the water is high, and vice versa.
5	**Water Flow** - Number shows the current rate of water flowing through the Unit, based on a flow signal (for variable flows) or a constant volume (for on/off steady flow applications).
6	**Power** - displays the current power level. The Unit's electronic ballast varies continuously (not in limited steps) to maintain the dose and respond to changing conditions.
7	**Dose Gauge**- represents the total system dose. The gauge has two different linear functions - from the dose setting down and from the dose setting up. The minimum and maximum allowable dosage levels are user-defined.
8	Press to access the System Status screen.

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Table 6-1: Main screen components

#	Item
9	Press to access Settings and Technician protected screens via Password entry screen.
10	**External Operation button** - appears on the main operating screen only if the External On/Off feature is activated in the main Settings screen. If the feature is enabled, the system can be turned on or off from the facility's External On/Off switching mechanism. When the feature is disabled, the system may be turned off only from this screen.
11	**Cleaning System button** - appears on the main operating screen only if theUltrasonic Cleaning feature is included in the system. Enables/disables the optionally-installedUltrasonic Cleaning mechanism for cleaning the Unit's quartz tubes.

6.1.1 Lamp status indication

The symbol next to each lamp number indicates the lamp status:

Table 6-2: Lamp status symbol

Symbol	Lamp Status
	(No symbol) Lamp Off
*	Lamp in Ignition
V	Lamp On
D	Lamp Disabled
E	Lamp Error

6.2 Planning System Configuration

Plan your Control Module configuration with your Atlantium Application Engineer to assure that the system performs properly and achieve its objectives. Review the Configuration Printout with the Atlantium Technician who is commissioning the site to make sure the settings are understood and correct. The Atlantium Technician configures the initial system settings via the Control Module. The User can afterwards change settings as experience suggests improvements in the operations and processes.

➲ **To access the Settings screen:**

Step 1. From the **Main Operations** screen, press [icon]. The **Password** screen appears.

Step 2. Using the numbered keys, enter the password (default password is **1357**) and press [icon] **OK**. The Settings screen appears.

Figure 6-3: Password Screen

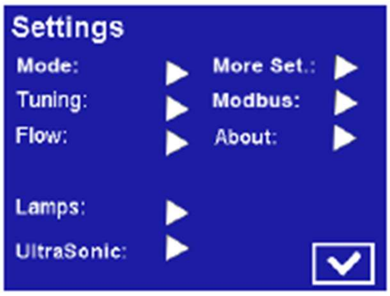


Figure 6-2: Settings Screen

6.3 Configuring the Operation Mode

The Control Module provides Atlantium system has two operational modes. Determine which suits your objectives and facility processes: of control:

- Power Mode - The system is set to operate at a particular power level. Under all circumstances, the system operates at the specified designated power level. Power mode is the right choice if maintaining a particular dose is not the objective. It is often chosen where energy consumption concerns are paramount or when the maximum power is needed at all times and concern that low flows might give high doses or that high flows might give lower doses is not an issue (i.e., maintaining a particular dose is not the objective).
- Dose Mode - The system is set to operate at a particular dose. It maintains the designated dose as conditions allow and generally is set to alarm when the dose is not achieved.

➲ **To set the Operation Mode:**

Step 1. Access the Control Module's Settings screen as detailed above and for **Mode**, press ▶. The Mode Setting screen appears

- Dose mode:
  - a For **Operation Mode**, use the ▲▼ arrows to select **Dose**.
  - b Ignore the **Power%** parameter
  - c To enable an alarm to be triggered should the Dose reach the minimum allowed, press the **Min Dos Alarm** box until a check-mark appears.

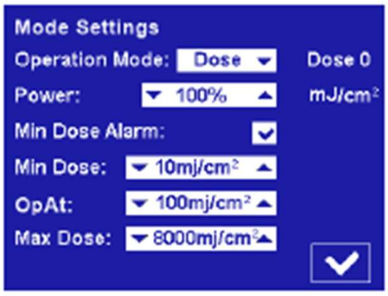


Figure 6-4: Dose Settings Screen

  - d For **Min Dose**, press the ▲▼ arrows to set the lowest allowable dose in mJ/cm^2. This sets the linear function for the left side of the dose gauge and is the dose the Unit maintains. If it is not maintained, it dips into the shaded area, and, if configured to send an alarm, it does so.

e For **Op**(erate) **at**, press the ▲▼ arrows, to set the operating dose. This generally includes a safety factor so that the Unit does not operate at the minimum margin and does not send false alarms when the water quality or flow quivers or changes rapidly.

f For **Max Dose**, press the ▲▼ arrows, to set the maximum UV dose limit. This sets the linear function for the right side of the gauge and defines when the arrow goes into the shady area. In cases where there is low flow, unusual water chemistry or the potential for overheating, high dose can indicate impending issues.

g Press OK. The settings are saved and you are returned to the Settings screen.

- Power mode:

a For **Operation Mode**, use the ▲▼ arrows to select **Pwr**.

b Set the **Power %** using the ▲▼ arrows.

c If you want to be alerted to the dose going beneath a value, set it here as the **Min Dose**. Press the ▲▼ arrows, to set the lowest acceptable dose in mJ/cm^2. This configures the left side of the gauge, but the system does not change the power or operation to achieve or maintain this dose. It is simply for information. If you set the **Minimum Dose alarm** (see below) it also triggers an alarm if the Unit falls below this dose.

d Ignore the **Operate at** parameter.

e For **Max Dose**, press the ▲▼ arrows, to set the maximum UV dose limit. This sets the linear function for the right side of the gauge.

f To enable the alarm trigger if the unit goes below the minimum allowed, press the **Min Dos Alarm** box until a check-mark appears.

g To save your changes and return to the Settings screen, press OK.

6.4 Configuring the Flow Settings

The Flow signal and related settings are critical for determining the dose and assuring that the system is working within design guidelines. The accuracy and tolerance of the flow meter measurement should match the Control Module interpretation of the flow signal.

➲ To configure the flow settings:

Step 1. Access the Control Module's Settings screen as detailed above.

Step 2. For **Flow**, press ▶. The Flow Settings screen appears.

Step 3. For **Flow Input**, use the ▲▼ arrows to select either **Meter** or **Switch**.

- If you selected **Switch**:
  - Mark the electronic polarity as **Normally Closed** or **Normally Open**.
  - The constant flow number is set by the Atlantium Technician in **Technician 2 Screen**.
- If you chose **Meter**:

a To set Gain, you need the meter's maximum-design flow rate (appears on the flow meter datasheet). You need to enter flow in m^3/h, so if it is written in any other unit of measurement, you must convert it to m^3/h.

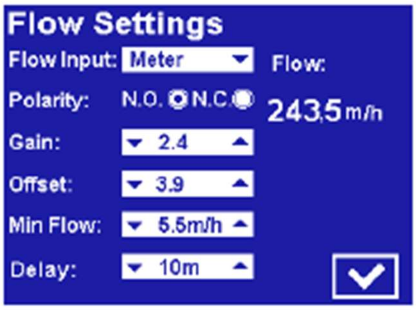


Figure 6-5: Flow Settings Screen

- For electronic polarity **Normally Closed** mark the NC check-box or for **Normally Open** leave the NC check-box blank.
- For regular wiring, set according to the table in ***External On/Off - Additional Notes*** on page 46

Step 8. To set the system to restart at the configured settings when the flow returns to normal rate, press the **Restart On Flow** check-box until a mark appears. When the check-mark is marked and the system shuts down because of low flow the main power remains on and a flow label on the main screen flashes indicating the system is waiting for a normal flow rate to resume). When normal flow rate returns, the system restarts and resumes operation according to the configured settings automatically.

Step 9. To save your changes and return to the Settings screen, press OK.

6.6 Configuring the Modbus

The Modbus screen displays the software version and other system version information.
Configure the Modbus node address assigned to this system in the Control Module. The Modbus enables communication with the rest of the facility and facilitates integration of the Atlantium Unit into the overall process.

➲ To configure the Modbus:

Step 1. Access the Control Module's Settings screen as detailed above.

Step 2. For **Modbus**, press ▶. The Modbus screen appears.

Step 3. For **Address**, using the ▲▼ arrows, set the Modbus address node to a number as per the serial address setting in your facility.

Step 4. For **BaudRate**, using the ▲▼ arrows, set the appropriate baud rate for this system. The options are:

- 115200
- 57600
- 38400
- 19200
- 9600

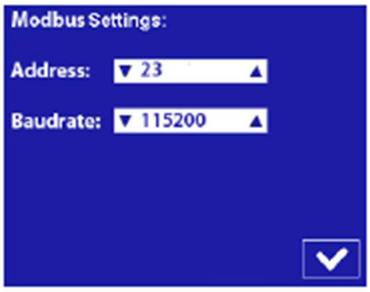


Figure 6-7: Modbus Settings Screen

Step 5. To save your changes and return to the Settings screen, press OK.

The software version and other system version data are displayed on the About screen.

Figure 6-8: About Screen

6.7 Configuring the Technician Settings

As a precautionary measure, the Technician settings in the Control Module should be verified.

➲ To configure the Technician Settings:

Step 1. From the **Main Operations** screen, press . The **Password** screen appears.

Step 2. Using the numbered keys, enter the password (default password is **8888**) for the Technician Screens and press OK. The **Technician 1** screen appears.

Step 3. For **System**, verify that the correct Atlantium system is defined.

Step 4. For **Auto On**, to set the Atlantium system to restart automatically when power is restored after a facility-wide power failure and reactivate according to the most recent settings, press the check-box until a mark appears.

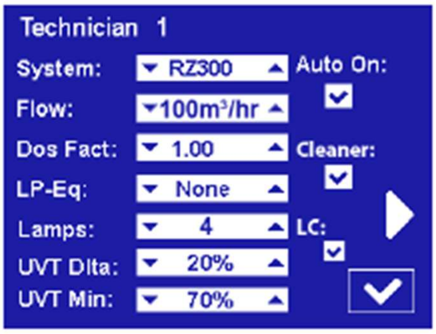


Figure 6-9: Generic Technician 1 Screen

Step 5. If the system includes an Ultrasonic cleaning system, press the **Cleaner** check-box until a check mark appears.

Step 6. If the system includes a **flow meter**, for **Flow**, set the maximum flow rate using the ▲▼ arrows. Otherwise, ignore this parameter.

Step 7. If the system uses a **flow switch** instead of a **flow meter**, in the **Flow** field, set the constant maximum flow rate using the ▲▼ arrows.

Be Careful when using this screen as changing the unit time or other errors can substantially impact your system!

Step 8. If your system contains PG quartz, the Dose Factor must be configured. For **Dos Fact**, set the Dose Factor using the ▲▼ arrows.

Step 9. For facilities that require Dose monitoring in Low Pressure Equivalence terms, for **LP-Eq**, use the ▲▼ arrows to set the Low Pressure Equivalence figure. The Dose is expressed as a Low Pressure Equivalent PDM system.

Step 10. Verify that the system is configured correctly for the number of lamps. Use the ▲▼ arrows to set the correct number.

Step 11. If the system is equipped with a special 10m long cable (applicable to RS104 only) check the **LC** check box

Step 12. To save your changes and return to the Settings screen, press ☑.

Step 13. To access the **Technician 2** screen, press the ▶ arrow on the right of the **Technician 1** screen.

Step 14. Verify that for each lamp, the parameters listed for UVT and UVI appear in the normal range.

Step 15. Ignore the **Power** option. It is used for testing only.

Step 16. Verify that for each lamp the column on the right is set to **ON** (enabling the lamps).

Step 17. To save your changes and return to the Settings screen, press ☑.

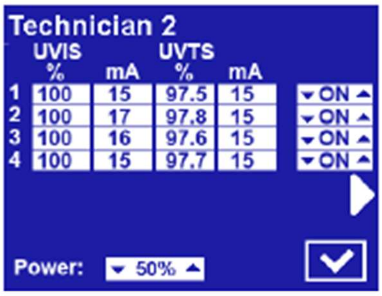


Figure 6-10: Technician 2 Screen*

Step 18. To access the **Technician 3** screen, press the ▶ arrow on the right of the **Technician 2** screen.

Step 19. Verify that the temperature displayed is within the norm. The temperature for each installed and operating lamp is displayed. An **N/A** appears for a lamp that is not operating or not installed.

Step 20. Ignore the **Power** option. It is used for testing only.

Step 21. Verify that for each lamp the column on the right is set to **ON** (enabling the lamps).

Step 22. If a lamp fails, it will be automatically disabled. The Volt column displays the failure code as received from the ballast PS (see **Section 6.7.1.** ***Lamp Failure codes in Technician screen 3***).

Step 23. To save your changes and return to the Settings screen, press ☑.

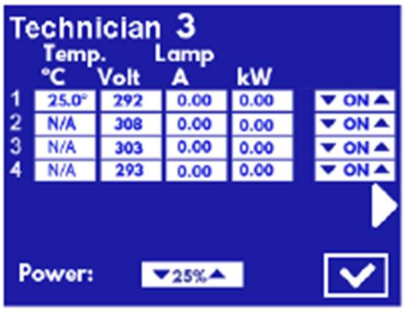


Figure 6-11: Technician 3 Screen**

*For a system with less than four lamps, only the relevant rows are displayed.
**For Temperature, for less than four lamps, only the fields for the installed lamps display the temperature. The other fields display N/A.

6.7.1 Lamp Failure codes in Technician screen 3

The table below provides the various "lamp off" or failure to ignite as identified by the ballast/s. Turn off reason can be identified in two ways:

- By a pop-up message
- By voltage value that acts as a **Failure code** and remains displayed in the voltage column in **Technician screen #3** after the lamp has gone off

Table 6-3: Table of Failure codes displayed in Technician screen #3

#Failure code	Reason for lamp turn off (appears in the pop-up message)
0 (+24)	None
50 (±24)	Lamp nominal voltage too low*
100 (±24)	Lamp nominal voltage too High**
150 (±24)	Device cooling insufficient
200 (±24)	Internal failure / One phase lost
250 (±24)	Lamp earth short circuit
300 (±24)	Lamp not ignited / gone out
350 (±24)	Lamp short circuit
400 (±24)	Phase failure mains voltage
450 (±24)	Mains over-voltage
500 (±24)	Manual turn off
600 (±50)	Loss of ballast's internal parameters***

** Voltage of lamp is low while the current to the lamp is at the maximum allowed value – in this state, the required power cannot be delivered to the lamp.*
*** The voltage required by the lamp cannot be created by the ballast. The current going to the lamp is zero.*
**** In case of loss of internal ballast parameters, consult with Atlantium service personnel for instructions on how to solve this issue.*

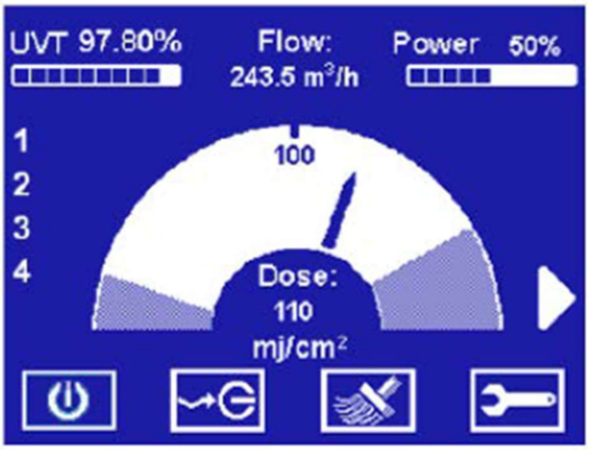
7 Initializing the Atlantium System

The Atlantium system starts up with the initialization of the Control Module.

➲ **To start up the Atlantium system:**

Step 1. Check that the power cables of the Ballast Module are all connected to the mains power source.

Step 2. To ignite the lamps, press the power button .



Step 3. Initial Main Operation Screen to ignite system.

8 First Time System Activation

Following Control Module configuration, completing the system commissioning includes the following major steps:

- Cleaning the Atlantium Unit (See below)
- Filling unit with water (See below)
- Igniting the UV Lamps (See below)
- Initiating the Water Flow (See page 8-85)
- Adjusting the Auxiliary Equipment (See page 8-85)
- System Tuning (See page 8-85)
- Installation QA (See page 8-86)

8.1 Cleaning the Atlantium Unit

The inner surfaces of the Atlantium Unit are to be cleaned via the CIP process at installation, before starting up and proceeding with the commissioning phase. Follow the directions in ***Cleaning In Place (CIP)*** on page 10-109.

8.2 Filling unit with water

The unit must be filled with water before full activation.

Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. **Make sure that water hammer is NOT present in your facility's water line.**

To prevent water hammer, consult your system engineer and check your facility system procedures, including:

- When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
- Sudden valve closures can cause water hammer. Make sure that valves close gradually enough to avoid this problem.
- Ensure that all pumps used with the Atlantium system employ soft start-up procedures.

➲ **To introduce water:**

Step 1. Open the Inlet valve and, using the Viewport on the Unit, check that the quartz chamber fills with water completely.

8.3 Igniting the Lamps

The lamps of the Atlantium Unit are ignited to commence live system operation.

- **Temporarily, the breather caps on the lamps must be left open to allow any condensation water to evaporate. During this time, set up Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- DO NOT look directly into openings that emit UV light.

To ignite the lamps:

Step 2. Post temporary Caution signs on the ground 1 m away from the Atlantium unit to warn bystanders of possible UV light exposure and require the use of appropriate skin and eye protection to approach.
Step 3. Open each of the lamp vent caps (2 on each side of the lamp). For the location of the lamp vent caps, see Figure 10-17 , on page 10-114.
Step 4. Check that the Main screen shows that system is working properly according to spec (see **Figure** 3)
Step 5. Press the power button and verify that power readings are as required.
Step 6. Press ▶ The System Settings screen appears.
Step 7. Configure system settings as required for your particular system. See ***Configuring the Control Module*** on page 6-74.
Step 8. Let the system operate for at least one hour.
Step 9. Continue with Section 8.5 below while the lamps complete the first hour of operation.
Step 10. After that first hour, close all of the lamps' breather caps and remove the Caution floor signs.

8.4 Initiating the Water Flow

When initiating the water flow, the water may be set to drain out or collect in a tank until the system is fully operational. Once the entire installation and QA testing is complete, the water is set to flow normally as part of the facility operation.

To initiate the water flow:

Step 1. Open the relevant valve that allows water to flow through the Atlantium unit. This may be the draining valve or bypass valve.
Step 2. Open the viewport and verify that the Unit is completely filled with water and that no air gaps and no air bubbles exist.
Step 3. From the **Main Operations** screen, press . The **Password** screen appears.
Step 4. Using the numbered keys, enter the password (default password is **1357**) and press OK. The Settings screen appears.
Step 5. For **Flow**, press ▶. The Flow Settings screen appears.
Step 6. Verify that the Flow reading on the right is within normal range. If it is not, check the connections between the Atlantium system and the Flow meter. See ***Connection Box*** on page 5-60.

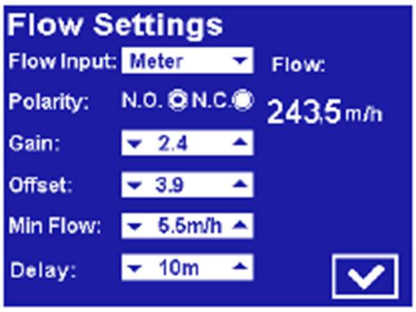


Figure 8-1: Flow Settings Screen

8.5 Adjusting the Auxiliary Equipment

All relevant auxiliary equipment, i.e., flow meter, automatic valves, PLC, etc., must be adjusted at this time. Refer to the manufacturers' user documentation of the relevant equipment.

If your configuration includes a flow meter, check that the value of the flow displayed on the Control Module matches the reading on the flow meter. See page 8-85 For flow meter calibration, see ***Configuring the Flow Settings*** on page 7-86.

8.6 System Tuning

The Atlantium system must be tuned via the Control Module. Follow the instructions in ***Replacing/Checking a UVT Analyzer or UVIS Sensor*** on page 10-129.

8.7 Installation QA

Atlantium personnel performs a quality assurance procedure (see ***Installation Check List*** on page 5-72) following which the Atlantium system is ready for regular facility operation.

9.3 Viewing System Status

You can view the real-time status of various System parameters.

➲ **To view the system's status:**

Step 1. On right side of the Control Module screen, press ▶. The System Status screen appears displaying the real-time status of the following parameters:

- UVT %
- Flow
- Power %
- Per Lamp: Power, Status, and Age
- Temperature

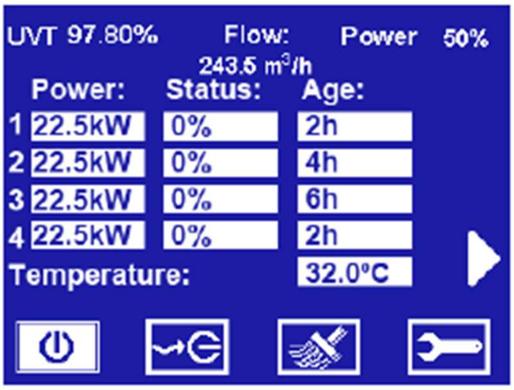


Figure 9-1: Control Module Status Screen

Every defined event is written to the event log of TRACS (if included). You can view the data log of all events in the **Events** tab of TRACS.